Task-Relevant Information Acquisition for Ultrasonic Inspection of Impacted Composites: From Spatial Information Structure to Evidence-Calibrated Sensing

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Abstract

Ultrasonic inspection resolves spatially rich internal morphology, but a cost-constrained mechanical decision need not acquire every observable detail. We introduce a two-part Task-Relevant Information Acquisition framework. Part I, Information Characterization, determines how measurement value is structured across space, sampling density, objective, inspection state, and downstream predictor. Part II, Evidence-Calibrated Decision Realization, converts that structure into legal partial-state valuation, matched information-source controls, bounded planning, and deployment calibration. The framework is evaluated on 276 specimens from six experimental domains using strict nested leave-one-domain-out validation and exact native-raster acquisition cost. A selected spatial field reduced equal-domain compression-after-impact prediction error by 32.1%, and a registered sparse condition retained 89.9% of the full-field gain. Retrospective mechanical and registered normalized-RGB-MSE reconstruction oracles selected task-specific priorities, while the teacher's best action changed for 70.4% of specimens as evidence accumulated. State-conditioned valuation improves over a frozen static ranking at the final registered checkpoint. Matched controls calibrate attribution, and controlled substitutions separate valuation from bounded set-planning headroom. The final frozen stress test quantifies a residual deployable gap of 6.114×10^{-5} in AUEBC relative to the strongest reference; the learned policy is favorable in two of six domains and is not performance-superior under this comparison. The resulting engineering-informatics contribution is an evidence-calibrated route from spatial information structure to a bounded sensing decision.

Keywords: adaptive ultrasonic inspection, compression after impact, value of information, task-driven sensing, held-out-domain evaluation

1. Introduction

Ultrasonic inspection produces spatially rich internal observations of impacted composites, but an engineering decision rarely needs every observable detail. Completed scans therefore contain more spatial information than residual- capacity assessment necessarily requires. For carbon-fiber-reinforced polymer (CFRP) structures, the acquisition problem is to identify which measurements justify their cost when distributed internal damage affects compression-after- impact (CAI) performance.

Full C-scan CAI prediction, sparse ultrasound, adaptive inspection, and task-driven sensing already establish important capabilities. Surface and image-resolved predictors support post-impact assessment [1, 2, 3, 4]; sparse reconstruction and adaptive paths reduce digital measurement burden [5, 6, 7]; and task-driven design and value-of-information methods connect sensing to downstream utility [8, 9, 10].

What remains is an integrated account of how task-relevant ultrasonic information is structured under limited sensing and how that structure should be converted into a causal, evidence-calibrated acquisition process. Spatial enrichment alone does not identify where limited measurement budget should be spent. Conditional value can also change as evidence accumulates, and a local value estimate must still be attributed to its information source and realized as a bounded sensing decision.

We address two progressive questions. RQ-A asks how task-relevant ultrasonic information is structured under limited sensing for downstream CAI assessment. RQ-B asks how this structure can be operationalized and evidence-calibrated for cost-constrained acquisition. Usefulness, task-value observability, and actionability are validation criteria within the two-stage framework rather than three independent success/failure programs.

The evidence program uses 276 paired specimens from six CFRP experimental domains, strict nested leave-one-domain-out evaluation, specimen-first inference, and exact native-raster location cost. Historical evidence and the outer policy endpoint remain frozen. Post-freeze diagnostics reuse hash-bound states and outcomes to characterize value, controls, components, planning, and feedback without modifying that endpoint. Retrospective teachers and oracles measure opportunity only and are never presented as deployable policies.

The contribution is threefold. First, Information Characterization provides a progressive account of spatial, sparse, heterogeneous, objective-conditioned, state-conditioned, and predictor-conditioned task value. Second, Evidence-Calibrated Decision Realization operationalizes that structure through a legal causal state, dynamic valuation, and matched controls under one causal acquisition contract; it is not a new generic definition of value of information. Third, controlled component and planning diagnostics followed by a frozen end-to-end stress test calibrate the remaining deployment gap without turning retrospective headroom into a superiority claim.

2. Related Research and Problem Formulation

2.1. Ultrasonic information for post-impact performance assessment

Low-velocity impact can leave limited visible evidence while producing a spatially distributed internal damage state that affects compression-after-impact (CAI) performance. Surface profiles have consequently been studied as accessible predictors of internal impact damage and residual CAI strength [1, 2]. These mappings establish that surface observations can be useful within a defined cohort, but they do not make the surface and internal fields informationally equivalent. Image-resolved analyses of impacted laminates likewise motivate retaining spatial morphology rather than reducing damage to a single projected extent [3].

The paired public releases used here provide specimen metadata, surface profiles, ultrasonic C-scans, and CAI measurements [11, 12]. Mack et al. subsequently showed that full C-scan images can support direct prediction of impact energy and CAI strength with a convolutional model [4]. That study is the closest application-level precedent: it establishes full-image predictive feasibility, so the present contribution is not the use of C-scans or image features for CAI regression. The unresolved engineering-information question is narrower and different. It asks which portions of the measured field carry task value, whether the value of an unacquired portion can be inferred from a legal partial inspection state, and whether that estimate improves the next measurement decision across held-out experimental domains.

This distinction also separates a predictive input from an acquisition policy. A full-field model may quantify the information available after the inspection has ended, but it does not tell an inspector which location to measure next or whether the value assigned to that location can be known before measurement. Those propositions require different information sets and different controls.

2.2. Sparse and adaptive ultrasonic inspection

Sparse ultrasonic sampling has been used to recover impact-delamination patterns from incomplete measurements [5]. Recent adaptive sampling of composite Lamb wavefields combines a damage-guided sampling pattern with a spatial–temporal masked autoencoder, with full-wavefield reconstruction as the learning endpoint [6]. Autonomous robotic ultrasonic inspection has also selected successive physical locations using Bayesian optimization over a damage-probability or novelty field [7]. Adaptive and sequential ultrasonic inspection are therefore established; neither is a priority claim of this work.

The acquisition objective matters. Reconstruction-oriented sampling rewards a measurement when it improves the recovered signal or image. Damage-localization sampling rewards evidence that resolves the presence or position of a defect. The present CAI task instead rewards measurements that reduce residual-capacity prediction error. These objectives can select the same locations, but this agreement cannot be assumed. A visually or numerically accurate reconstruction may preserve features irrelevant to CAI, whereas a mechanically useful measurement need not minimize global image error. We therefore compare mechanics- and reconstruction-conditioned utilities under a common acquisition and cost contract.

Sparse fractions in this paper refer to locations on a registered digital raster. They are not converted into scanner time. Physical travel, coupling, settling, pulse repetition, and path-planning overhead are absent from the released data, so a native-raster measurement reduction is not evidence of an equal inspection-time reduction.

2.3. Task-driven sensing and value of information

Task-driven experimental design explicitly chooses measurements for a downstream objective rather than for signal reconstruction alone. For example, TADRED selects a compact global channel design from dense multi-channel images while optimizing a user-specified task [8]. In ultrasonic structural health monitoring, value-of-information criteria have been used to trade diagnostic information against the number and placement cost of guided-wave sensors [10]. These works establish both task-driven imaging and ultrasonic value-of-information design as prior art.

Sequential decision making adds a further dependency: the value of a future observation depends on what is already known. In a partially observable infrastructure-management setting, component inspection, monitoring, and system-level scheduling are distinct decisions because beliefs and future opportunities evolve after observations [9]. This separation is directly relevant to partial C-scan inspection. A retrospective calculation may show that a location would have reduced CAI error, while a deployable score may fail to identify that location from the current state; even an accurate one-step score may fail when measurements must be selected as a cost-constrained set.

Accordingly, this paper does not present task-driven acquisition itself as the novelty. Its contribution is an operational progression from Information Characterization to Evidence-Calibrated Decision Realization under a causal partial-state contract. Usefulness, task-value observability, and actionability serve as validation criteria within that progression.

Table 1 positions this contribution against the six closest sources without treating the validation labels as the novelty.

2.4. Problem formulation and research questions

Let specimen i belong to experimental domain $d(i)$. Its response y_i is the damaged-to-intact CAI-strength ratio. Let z_i denote legal pre-inspection context, S_i the complete native-raster ultrasonic field, and $\mathcal{I}_{i,t}$ the information legally available after acquisition step t. The latter contains z_i, acquired coordinates and values, action history, and exact cost, but excludes unrevealed values in S_i and the outcome y_i. A candidate measurement $X \in \mathcal{A}(\mathcal{I}_{i,t})$ is legal only if it refines the current hierarchical measurement state without exceeding the budget.

For a fixed downstream predictor f and task loss ℓ, define the risk of a state as

$$\mathcal{R}_f(\mathcal{I}_{i,t}) = \ell(y_i, f(\mathcal{I}_{i,t}))\,. \tag{1}$$

The realized task value of candidate X is

$$U_f(X \mid \mathcal{I}_{i,t}) = \mathcal{R}_f(\mathcal{I}_{i,t}) - \mathcal{R}_f(\mathcal{I}_{i,t} \cup X). \tag{2}$$

A positive value means that revealing X reduces error for this predictor at this state. Because both terms depend on f, Eq. (2) defines *downstream-predictor-conditioned task value*; it is not an intrinsic or universal mechanical property of a location. During retrospective analysis, U_f may use the true

Table 1: Closest-work positioning by measurement setting, acquisition objective, state dependence, downstream endpoint, and scope. The present study connects information characterization to evidence-calibrated decision realization under one causal acquisition contract; it does not claim the first adaptive ultrasonic, ultrasound-VoI, or task-driven design.

Work	Measurement	Objective	State dependence	Endpoint	Outside scope
Fuentes et al. (2020) [7]	robotic ultrasonic A-scan features sampled over a C-scan surface	Bayesian-optimization search over novelty and damage probability	sequential posterior update and next-location selection	damage and novelty localization	CAI-conditioned value and held-out-domain closed-loop utility were not study objectives
Cantero-Chinchilla et al. (2020) [10]	ultrasonic guided-wave structural-monitoring sensors	expected value of information with sensor-number and placement cost	static sensor configuration	Bayesian damage diagnosis and localization	within-specimen partial C-scan reveal and CAI prediction were not study objectives
Memarzadeh and Pozzi (2016) [9]	probabilistic component inspection and permanent monitoring	value of information for inspection and system scheduling	sequential belief and decision updates	maintenance and inspection scheduling	ultrasonic imaging and CAI mechanics were not study objectives
Blumberg et al. (2024; TADRED) [8]	densely sampled multi-channel imaging followed by sparse channel design	task-driven feature and channel selection	static global subset design	user-specified downstream image-analysis task	state-dependent within-specimen causal reveal and exact-cost sequential planning were outside scope
Ji et al. (2026) [6]	sparse SLDV Lamb wavefield measurements on composite structures	Bayesian-optimization-guided sampling for STMAE reconstruction	adaptive spatial sampling-pattern construction	full wavefield reconstruction and damage-area fidelity	CAI mechanics and predictor-conditioned acquisition value were not study objectives
Mack et al. (2026) [4]	complete ultrasonic C-scan images of impacted composites	no acquisition objective; full scans are inputs	static full-field prediction	impact energy and CAI-strength regression	sparse or adaptive acquisition and conditional next-measurement value were outside scope

response and counterfactual revealed measurements. Such values are teacher or oracle evidence and are unavailable to a deployable policy.

The study addresses two progressive research questions:

RQ-A — Information characterization: How is task-relevant ultrasonic information structured under limited sensing for downstream CAI assessment?

RQ-B — Decision realization: How can this task-relevant information structure be operationalized and evidence-calibrated for cost-constrained acquisition?

Usefulness, task-value observability, and actionability are validation criteria within the two-stage framework. They test predictive enrichment, legal-state valuation, and decision consequence without imposing three independent result programs.

3. Task-Relevant Information Acquisition Framework

3.1. Framework overview

Task-Relevant Information Acquisition combines two scientific parts (Fig. 1). Part I, Information Characterization, asks how ultrasonic information is organized across space, sparse observation, candidate heterogeneity, downstream objective, inspection state, and predictor. These dependencies produce the construct $U_f(X \mid \mathcal{I}_{i,t})$. Part II, Evidence-Calibrated Decision Realization, uses a legal causal state to estimate that construct, tests its information source with matched controls, separates valuation from bounded planning, and calibrates the complete deployment-facing trajectory.

The two parts are progressive but evidentially non-equivalent. A retrospective oracle can characterize where opportunity exists without defining a policy; an accurate local value estimate still has to be attributed to legal information and converted into a bounded action set. Here, "does not imply" is an evidential rule about licensed claims, not an information-theoretic statement.

3.2. Characterizing task-relevant information

Information Characterization changes one dependency at a time while holding the cohort, held-out domains, response, and validation logic fixed.

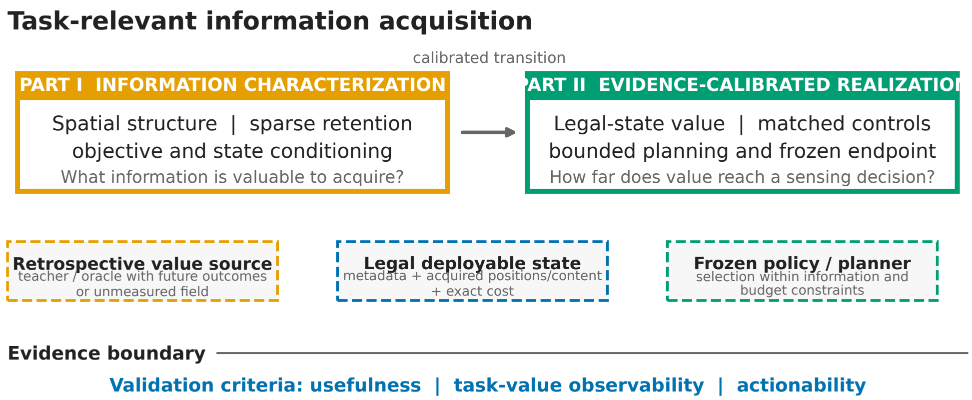

Figure 1: Task-Relevant Information Acquisition framework. Part I characterizes spatial, sparse, heterogeneous, objective-, state-, and predictor-conditioned information and yields $U_f(X \mid \mathcal{I}_{i,t})$. Part II realizes that construct through legal-state valuation, matched controls, component and set-level diagnostics, and frozen deployment calibration. Usefulness, task-value observability, and actionability annotate validation criteria rather than forming a success ladder.

Matched scalar-to-spatial comparisons test morphology enrichment; a registered sparse protocol tests recoverability; counterfactual candidate designs test spatial heterogeneity; common-cost mechanical and reconstruction oracles test objective conditioning; causal histories test state conditioning; and repeated values under several predictors test the index f.

For candidate designs, the retrospective oracle chooses

$$X^{*}_{f,i,t} = \arg\max_{X \in \mathcal{A}(\mathcal{I}_{i,t})} U_f(X \mid \mathcal{I}_{i,t}) \quad \text{subject to} \quad c(X \mid \mathcal{I}_{i,t}) \leq b_t, \tag{3}$$

where c is exact added native-raster cost and b_t is the remaining budget. Evaluation requires counterfactual measurements and y_i, so the oracle measures headroom and remains non-deployable. Reconstruction comparisons are restricted to the registered normalized-RGB-MSE reconstruction objective; they do not generalize to every image metric.

3.3. Causal partial-state valuation

Let g_θ be a scorer restricted to the legal current state and candidate descriptor. Its prediction is

$$\widehat{U}_{i,t,X} = g_\theta(\mathcal{I}_{i,t}, X, c(X \mid \mathcal{I}_{i,t}))\,. \tag{4}$$

For each outer target domain, teacher predictors, scorer fitting, and selection use source domains only. A strict-OOF retrospective teacher evaluates each legal candidate through

$$u_{i,t,X} = |y_i - \widehat{y}_i(\mathcal{I}_{i,t})| - |y_i - \widehat{y}_i(\mathcal{I}_{i,t} \cup X)| \,. \tag{5}$$

The held-out domain is queried only after fitting and selection are fixed. Teacher evolution characterizes the moving target; a deployable dynamic scorer is then evaluated against that target with candidate ranking, best-action and top-k agreement, and exact-budget set regret.

3.4. Evidence calibration through matched controls

Four controls determine which information-source claims are licensed. A static reference removes partial-state conditioning. The acquired-position/history control inherits the exact acquired coordinates and cost of each frozen state while removing measurement values; because histories arise from random, uniform, reconstruction-driven, and oracle-derived trajectories, it does not isolate geometry alone. A reconstruction-derived control tests whether a legal partial field supports prediction through image recovery. A shuffled- content control preserves recipient positions and budget while replacing measured values with source-separated donor content.

These controls separate response to an inspection history from evidence attributable to specimen-specific measured values. Improvement over the static reference licenses state dependence. Attribution to measured content requires the real-state representation to survive the acquired-position/history, reconstruction-derived, and shuffled-content comparisons.

3.5. Decision realization and deployment calibration

At state $\mathcal{I}_{i,t}$, a deployable selector chooses a legal candidate using current information and exact incremental cost. A one-step value-per-cost rule is

$$\pi(\mathcal{I}_{i,t}) = \arg \max_{X \in \mathcal{A}(\mathcal{I}_{i,t})} \frac{\widehat{U}_{i,t,X}}{c(X \mid \mathcal{I}_{i,t})}. \tag{6}$$

The frozen testbed also evaluates bounded lookahead, but every choice obeys the same reveal and budget contract. For effective specimen budgets $x_{i,1} < \cdots < x_{i,K}$ with CAI errors $e_i(x_{i,k})$, policy quality is summarized as

$$\mathrm{AUEBC}_i = \frac{1}{x_{i,K} - x_{i,1}} \sum_{k=1}^{K-1} \frac{e_i(x_{i,k}) + e_i(x_{i,k+1})}{2} (x_{i,k+1} - x_{i,k}), \tag{7}$$

the budget-span-normalized trapezoidal mean error over the observed effective-budget range, where x is the actual/effective specimen budget and lower values are preferred.

MAVIS is used as a frozen operational testbed for this calibration, not as a superior adaptive scanner. Its policy context contains metadata and surface/context features but no complete scan or CAI outcome; a separate source-teacher view exposes privileged fields only for source-domain teacher construction and retrospective evaluation. Causal reveal begins with a geometry-neutral uniform scout. The native raster is partitioned into 64 cells with three refinement levels, and only coordinates implied by legal actions are materialized. Revealed coordinates and values, measurement levels, action history, native shape, and exact acquired count form immutable states.

The current-state encoder aggregates variable-length measured tokens and fuses them with context and exact-cost features. The dynamic scorer receives this state plus candidate geometry, level transition, added cost, and remaining cost. Rollout reveals only registered coordinates and either re-scores after each update or retains the initial ordering for the no-feedback reference. Retrospective substitutions and bounded set diagnostics localize remaining headroom; feedback and the frozen outer comparison calibrate whether that headroom becomes end-to-end deployment-facing benefit.

4. Multi-Domain CFRP Case Study and Experimental Design

4.1. Specimens and ultrasonic/CAI measurements

The case study uses 276 CAI-complete carbon-fiber-reinforced polymer specimens from six released experimental datasets [11, 12]. The domains contain 45, 49, 43, 59, 42, and 38 specimens, respectively. They differ in laminate configuration and experimental conditions and are retained as indivisible validation units. The response is the damaged-to-intact CAI-strength ratio. Available modalities are specimen metadata, surface observables, an ultrasonic RGB C-scan of the internal damage field, and the CAI outcome.

The registered scan manifest binds every specimen identifier to its domain, native raster shape, pixel count, and source-content hashes. Native rasters are not resized for acquisition-cost accounting. This matters because scan sizes vary across the cohort: a nominal percentage alone does not specify the number of newly observed locations. Table 2 summarizes the cohort and evaluation contract.

Table 2: Multi-domain CFRP case study and evaluation protocol.

Item	Protocol value
Physical cohort	276 CAI-complete physical specimens
Held-out domains	6 experimental domains
Domain specimen counts	74t7kcdgkr: 45; cgtnjyggtm: 49; w68dtmpfyf: 43; xcmzfsbd9t: 59; yfxyg8jm46: 42; ykhs7s2dck: 38
Available modalities	specimen metadata; surface observables; ultrasonic RGB C-scan internal field; CAI-ratio outcome
Validation protocol	strict nested leave-one-domain-out (LODO); outer-domain outcomes excluded from training and selection
Registered checkpoints	3.125%, 6.25%, 9.375%, 12.5%, 18.75%, 25%
Acquisition cost	exact unique newly observed native-raster locations divided by native count; no scanner-time equivalence
Teacher/oracle information	future outcomes, unmeasured fields, or counterfactual utilities; retrospective and non-deployable
Deployable information	specimen metadata, acquired positions, measured content, legal actions, and current exact cost
Statistical units	physical specimen first; held-out domain second; six domains receive equal weight
Bootstrap aggregation	specimens resampled within domain with synchronized contrasts; domain means then equally weighted
Computational rows	state-action and checkpoint rows are repeated computational records, not independent statistical samples

4.2. Information representations and candidate measurements

The usefulness experiments compare three information resolutions. The scalar reference combines metadata, surface observations, and scalar summaries of the internal scan. The matched spatial candidate uses the same registered predictor family but retains the selected full C-scan field representation. A separate internal-only field estimator is reported only as a sensitivity path; it is not substituted for the matched confirmatory comparison. The sparse retrospective condition reveals a registered 25% sampling pattern on a normalized raster and uses bilinear interpolation before applying the frozen field pathway.

For sequential acquisition, each native raster is divided into an 8×8 cell grid. A cell may move through three registered sampling levels, and an action refines exactly one cell by one level. Candidate descriptors contain the cell position, level transition, exact number of newly observed native locations, native count, and remaining exact budget. This representation permits comparison of uniform, random, reconstruction-conditioned, mechanics-conditioned, static, and state-conditioned choices without changing the legal action space.

Mechanics utility is the change in CAI prediction loss in Eq. (2). Reconstruction utility uses normalized RGB mean squared error over the same registered raster. These utilities are kept separate: an acquisition may be optimal for one and suboptimal for the other.

4.3. Held-out-domain evaluation protocol

All headline results use leave-one-domain-out (LODO) evaluation. Each of the six experimental datasets serves once as the outer target domain. Model fitting, normalization, hyperparameter or checkpoint selection, teacher construction, policy choice, and fallback thresholds use only the other five source domains. The outer-domain outcomes are excluded until the frozen candidate is evaluated. When an inner selection loop is required, entire source domains remain intact. This nested structure avoids the optimistic error estimates that arise when model selection and assessment reuse observations [13] and aligns the validation unit with the intended unseen-experiment question rather than a specimen-random split [14].

The physical specimen is the primary statistical unit. Multiple candidate actions, random seeds, and checkpoints for one specimen are repeated computational records, not independent samples. Domain effects are first computed from paired specimen outcomes inside each held-out domain and then aggregated with equal weight across the six domains.

4.4. Privileged, deployable, and forbidden information

The information boundary is defined by when and where a variable is available. Metadata and surface observables are legal before ultrasonic acquisition. Acquired positions, measured RGB values, measurement levels, legal actions, and exact current cost are deployable state variables. Reconstructions from that legal state may be used as controls. Unacquired ultrasonic content and the true CAI outcome are forbidden to a deployable scorer or policy.

Teacher and oracle analyses deliberately use privileged information to define retrospective targets and headroom. Their outputs are never described as deployable. In particular, a true-value oracle may inspect counterfactual outcomes for all legal actions, and a joint near-oracle may compare reachable action sets. These rows answer whether an opportunity exists or where a bounded decision gap lies; they do not specify an implementable inspector.

4.5. Exact acquisition cost, metrics, and statistical inference

The registered checkpoints are 3.125%, 6.25%, 9.375%, 12.5%, 18.75%, and 25% of the native raster. Initial scouts are 1.5625% except in the one domain whose native geometry requires a 3.125% scout. At every transition, cost is the number of unique newly revealed native-raster coordinates. Effective budget is that count divided by the specimen's native pixel count. Previously observed coordinates have zero added cost, and an action that would exceed the current cap is illegal. No result is translated into physical scanner time.

For full-field and sparse prediction, the primary outcome is equal-domain mean absolute error in CAI ratio. Sequential CAI performance is summarized by AUEBC from Eq. (7); lower values are better. Reconstruction is measured by normalized RGB mean squared error. Observability metrics include Spearman rank agreement between predicted and strict-OOF teacher action values, best-action turnover or agreement, top-k overlap, and exact-budget set regret. Actionability contrasts use paired differences in specimen AUEBC.

Uncertainty intervals use synchronized bootstrap resampling of physical specimens within each held-out domain. The same resampled indices are applied to both methods in a contrast, domain means are recomputed, and the six domain means are equally weighted. The registered scalar-to-field family uses its predeclared familywise simultaneous interval. Other intervals retain the registered paired-bootstrap definitions. With only six domains, domain counts are always reported beside the aggregate effect rather than treated as a large-sample inferential substitute; this is consistent with the caution required for clustered data with few groups [15].

4.6. Validation criteria and evidence chronology

RQ-A is evaluated through matched spatial enrichment, sparse recoverability, retrospective spatial heterogeneity, cross-objective specificity, conditional teacher evolution, and predictor sensitivity. RQ-B begins with the static reference, evaluates dynamic state-conditioned valuation, calibrates information-source attribution with matched controls, separates valuation from bounded planning, and closes with feedback and the frozen policy comparison.

Usefulness validates predictive task enrichment, task-value observability validates inference from legal state, and actionability validates decision consequence. These criteria prevent evidence migration between stages. Oracle headroom cannot support a deployable policy claim; state dependence cannot support content attribution unless matched controls license it; and a planning diagnostic cannot support improved acquisition without the frozen end-to-end comparison. Section 5 reports the two progressive parts in this order.

The evidence chain contains distinct chronological roles. Historical comparisons and the final outer policy endpoint were frozen before the post-freeze science-closure diagnostics. Later analyses reuse hash-bound frozen states and outcomes to interrogate value evolution, state controls, task specificity, learner stability, component attribution, planning, and feedback. They were not used to re-select or modify the frozen outer endpoint.

5. Experimental Results and Discussion

5.1. Part I — From Spatial Morphology to State-Conditioned Task Value

5.1.1. Spatial morphology enriches residual-capacity information

The registered matched comparison supported a task-usefulness gain from spatial internal morphology. Equal-domain CAI-ratio MAE was 0.18920 for the scalar reference and 0.12849 for the selected spatial field, a reduction of 0.06072 (32.1%; familywise 95% CI 0.00664–0.15364). The field reduced error in five of six held-out domains. Relative to the metadata-and-surface reference, whose MAE was 0.18812, the same selected field reduced MAE by 0.05963 (31.7%; familywise 95% CI 0.00722–0.14842), again improving five domains. Thus the result is not an artifact of selecting an unusually weak scalar summary: both registered references lead to the same bounded conclusion that the spatial internal field contains CAI-relevant information absent from the lower-dimensional inputs.

A distinct internal-only field estimator produced a lower point MAE of 0.08964. It is retained as a sensitivity estimate, not substituted for the matched confirmatory path. The registered effect is therefore the 0.06072 reduction, not a cross-family difference constructed from the lowest available point estimate. Figure 2a shows the matched comparison.

5.1.2. Sparse observations preserve most task-relevant information

The selected 25% bilinear sparse condition had MAE 0.13451. Relative to the surface reference, its reduction was 0.05361 (95% CI 0.00561–0.13544), with lower error in five of six domains. This condition retained 89.9% of the registered full-field gain. Retention was substantial but incomplete: sparse MAE remained 0.00602 above the selected full field (95% CI 0.00173–0.01083), and the sparse condition was worse in every held-out domain. The appropriate engineering conclusion is therefore that a registered sparse digital measurement retains most of the full-field task information, not that the sparse and full fields are equivalent or that physical scan time falls by the same percentage (Fig. 2b).

5.1.3. Measurement utility is spatially heterogeneous

Retrospective oracle comparisons showed that the useful information was spatially heterogeneous. One-shot mechanical-oracle CAI AUEBC was 0.013458, compared with 0.017363 for uniform acquisition and 0.017188 for the one-shot reconstruction oracle. The corresponding improvements were 0.00391 (95% CI 0.00280–0.00502) and 0.00373 (95% CI 0.00284–0.00469), respectively, and both contrasts favored the mechanical oracle in all six domains. The one-shot oracle retained 56.8% of the registered sequential-oracle headroom. These values establish retrospective specimen-specific opportunity, but the oracle uses unavailable counterfactual outcomes and is not a policy.

5.1.4. Downstream objectives induce distinct measurement priorities

Under the common exact-cost trajectory contract, oracle acquisition depended on the objective. The mechanics-optimal oracle improved CAI AUEBC relative to the reconstruction oracle by 0.04862 (95% CI 0.04527–0.05205). In the opposite direction, the reconstruction-optimal oracle improved normalized RGB image MSE by 5.503×10^{-4} (95% CI 5.006×10^{-4}–6.063×10^{-4}). The registered normalized-RGB-MSE reconstruction objective and the mechanics objective therefore induce distinct retrospective priorities (Fig. 2c).

The learned global-mask control bounds this conclusion. Source-trained global mechanics and reconstruction masks produced a predeclared cross-objective support indicator of 0: the learned global mechanics mask does not reproduce the oracle separation. The result supports objective-conditioned opportunity, not learned deployment of that distinction.

5.1.5. Measurement value evolves with inspection state

The strict-OOF retrospective teacher changed as legal information accumulated. Between the initial state and the 18.75% checkpoint, the best action changed for 70.4% of specimens. Spearman agreement with initial action values fell to 0.405, top-five Jaccard overlap was 0.307, and the descriptive dynamic-versus- initial opportunity was 0.00531. These specimen-first diagnostics cover all 276 specimens across the six held-out domains and demonstrate

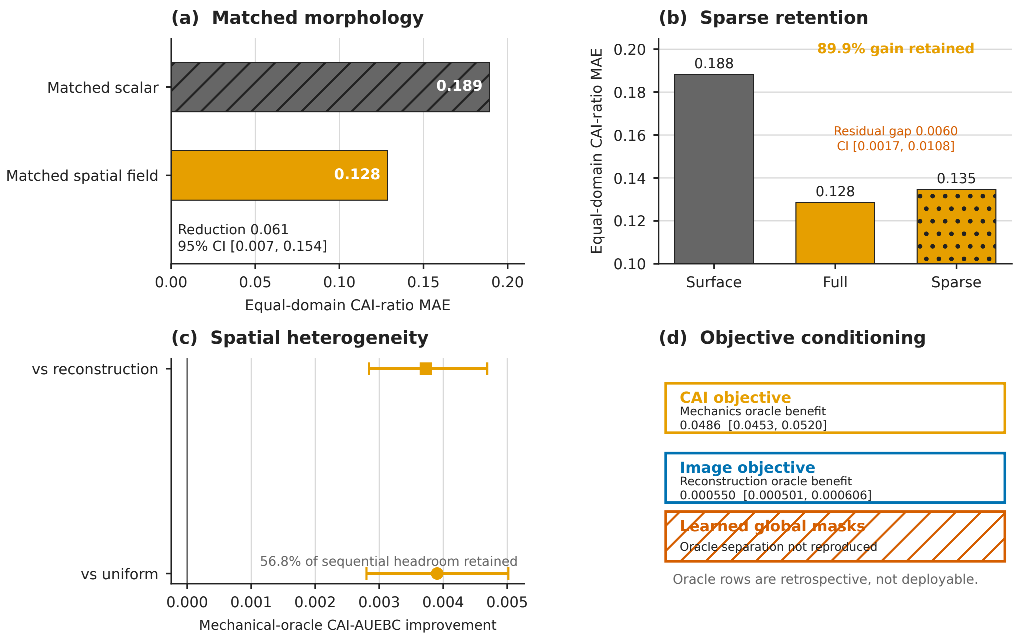

Figure 2: Part I: information characterization. (a) Registered spatial morphology enriches CAI-ratio prediction relative to matched scalar input. (b) Sparse observations retain 89.9% of the registered full-field gain while remaining distinct from the full field. (c) Retrospective mechanical value is spatially heterogeneous. (d) Mechanical and reconstruction objectives induce distinct oracle priorities, while learned global masks do not reproduce that separation. Oracle and sparse-design comparisons do not establish deployability or physical scanner-time savings.

that task value is state-conditioned rather than a fixed coordinate ranking (Fig. 2d).

This evolution belongs to the retrospective teacher, not automatically to a deployable scorer. It motivates state-conditioned valuation while requiring Part II to establish which legal-state information drives the decision signal.

5.1.6. Predictor conditioning bounds the task-value definition

Action-value maps were relatively stable between the two comparably performing low-complexity predictors and did not persist under the substantially less accurate shallow MLP. Full-state equal-domain strict-OOF MAE was 0.08964 for Ridge, 0.08618 for Huber, and 0.15067 for the shallow MLP. Ridge and Huber strict-OOF values had Spearman agreement 0.762 (95% CI 0.699–0.821), best-action agreement 0.675 (95% CI 0.622–0.727), and top-k

Jaccard agreement 0.685 (95% CI 0.650–0.719). Ridge and shallow-MLP values instead had rank agreement 0.116 (95% CI 0.069–0.164), best-action agreement 0.233 (95% CI 0.182–0.283), and top-k agreement 0.213 (95% CI 0.192–0.235). We therefore retain the predictor index f in U_f and avoid treating the value map as predictor-invariant. The current experiment does not determine how much value-map variation remains among equally accurate but structurally distinct predictors.

Part I establishes a structured task-relevant information target: spatial morphology improves residual-capacity prediction, sparse observations preserve most of that gain, mechanical opportunity varies across locations and objectives, and its ranking evolves with state. The target remains explicitly retrospective, downstream-predictor-conditioned, and bounded by the learned global-mask and predictor-sensitivity controls.

5.2. Part II — From State-Conditioned Value to Evidence-Calibrated Decisions

5.2.1. A static reference motivates state-conditioned valuation

The registered static scorer did not rank strict-OOF action values across held-out domains. Its equal-domain Spearman correlation was -0.0196 (95% CI -0.0591–0.0195), with a favorable direction in three of six domains (Fig. 3a). Exact-budget set regret was 0.08171 for the static scorer, compared with 0.07993 for the global mechanical score and 0.07978 for the random median. Increasing static scorer complexity was not authorized by these results: the registered representation had not first established specimen-specific content value across the six held-out domains.

The confidence interval spans zero, so this result does not prove that value is information-theoretically unobservable. It shows that the tested pre-acquisition information and scorer did not recover the strict-OOF target under whole-domain holdout.

5.2.2. Dynamic valuation captures incremental state dependence

At the 18.75% endpoint, the dynamic real-state scorer reduced one-step value regret relative to the static scorer by 0.00126: dynamic minus static was -0.001260 (95% CI -0.002123 to -0.000444), favorable in five of six domains. This narrow positive result establishes incremental state dependence relative to the static reference, while the next stage determines which source of state information licenses that interpretation.

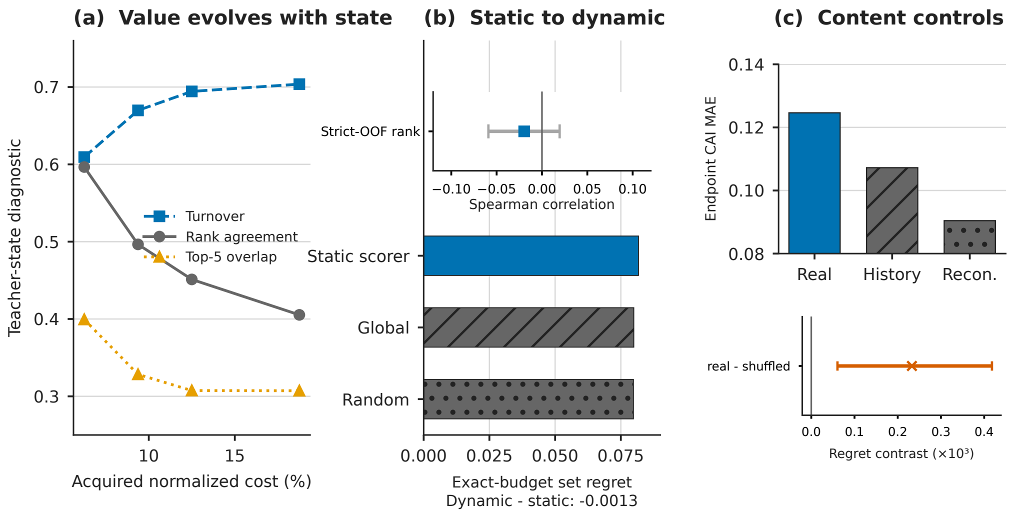

Figure 3: State-conditioned valuation and source controls. (a) Strict-OOF teacher values evolve with acquisition state. (b) Dynamic valuation improves one-step regret relative to the static reference. (c) Real partial measurements change prediction from the initial state, while acquired-position/history, registered reconstruction, and shuffled-content controls delimit content attribution. Intervals use synchronized specimen-bootstrap contrasts where shown.

5.2.3. Control decomposition identifies the source of decision signal

Real partial measurements changed CAI prediction relative to the initial state. At 25%, real-state MAE changed by −0.000731 (95% CI −0.001184 to −0.000250), and the endpoint diagnostic recovered 26.6% (95% CI 10.6%–41.0%) of the static-to-independent-full-field error ratio. That denominator is the separate internal-only full-field sensitivity estimator, not the matched spatial estimator used in Part I.

Matched controls localize the evidential boundary. At 25%, real-content MAE was 0.12463, acquired-position/history control MAE was 0.10722, and reconstruction control MAE was 0.09043; real minus acquired-position/history was 0.01740 (95% CI 0.00890–0.02582), and real minus reconstruction was 0.03419 (95% CI 0.02450–0.04384); real content was favorable in one of six domains for each. The positions-only input inherits the acquired-position history from the frozen state and is therefore an acquired-position/history control, not a pure geometry-only control. The reconstruction input follows the registered normalized-RGB-MSE reconstruction objective.

The dynamic source control gives the same boundary: dynamic real minus shuffled regret was 2.328×10^{-4} (95% CI 6.060×10^{-5}–4.180×10^{-4}), favorable for real content in one of six domains. The real-state change is genuine, but these matched controls do not license specimen-specific measured-content attribution for the registered scorer; they do not claim that acquisition history or reconstruction is universally sufficient.

5.2.4. Component attribution separates valuation and planning headroom

Retrospective substitutions separated valuation from bounded planning effects (Fig. 4a). Replacing learned valuation with privileged retrospective values improved CAI AUEBC by 4.979×10^{-5} (95% CI 1.845×10^{-5}–8.105×10^{-5}). A bounded learned-planning substitution improved AUEBC by 3.164×10^{-6} (95% CI 1.411×10^{-7}–6.250×10^{-6}), whereas stronger planning with true values improved it by 1.117×10^{-4} (95% CI 9.898×10^{-5}–1.248×10^{-4}).

These contrasts show that valuation and planning can contribute distinct shortfalls. They do not prove that either retrospective substitution is deployable. A separate representation-bottleneck row was not interpreted because the matched observability controls in Section 5.2.3 did not establish content-specific state value.

5.2.5. Bounded set realization exposes decision-level headroom

Within the predeclared two-action reachable pool, current greedy selection had regret 1.207×10^{-4} relative to a retrospective joint near-oracle set (95% CI 1.033×10^{-4}–1.386×10^{-4}). Beam search with width four reduced the point regret to 1.127×10^{-4}, but its interval remained positive (95% CI 9.485×10^{-5}–1.308×10^{-4}). The diagnostic therefore supports a bounded set-planning gap (Fig. 4b). It neither makes the joint near-oracle deployable nor establishes arbitrary-horizon planning regret.

5.2.6. End-to-end stress testing calibrates deployment readiness

The no-feedback reference retained a CAI AUEBC advantage of 1.496×10^{-5} (95% CI 1.064×10^{-5}–1.944×10^{-5}) under the frozen implementation. Equivalently, with the registered sign convention of no-feedback error minus feedback error, feedback benefit was -1.496×10^{-5} (95% CI -1.944×10^{-5} to -1.064×10^{-5}), and feedback was favorable in two of six held-out domains. The stress test retains the simpler no-feedback reference and does not generalize the direction to other representations or policies.

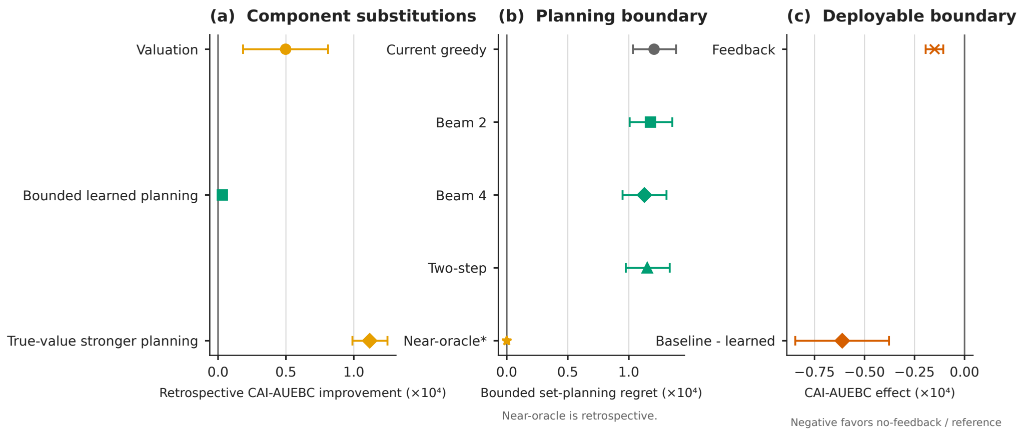

Figure 4: Decision realization and deployment calibration. (a) Controlled substitutions expose distinct valuation and bounded planning headroom. (b) Reachable-pool analysis quantifies a positive set-realization gap while retaining the near-oracle as non-deployable. (c) The no-feedback reference is retained by the frozen stress test, and the strongest deployable reference retains a residual AUEBC advantage, so the learned endpoint is not performance-superior.

The frozen learned policy had equal-domain CAI AUEBC 0.125053, compared with 0.124992 for the strongest deployable baseline. Baseline minus learned-policy AUEBC was -6.114×10^{-5} (95% CI -8.461×10^{-5} to -3.777×10^{-5}), and the learned policy improved two of six held-out domains. The residual deployable gap is therefore 6.114×10^{-5} in favor of the reference. The learned endpoint is not performance-superior under the frozen comparison. This calibration coexists with the positive retrospective valuation and planning headroom; it identifies the remaining representation-to-decision gap rather than erasing that upstream structure.

Part II shows that state conditioning improves over the static reference, matched controls determine which information-source claims are licensed, controlled substitutions expose valuation and set-planning headroom, and the final frozen stress test quantifies the residual deployment gap.

5.3. Integrated interpretation

Table 3 assembles the two-part evidence chain while preserving the validation criterion, chronology, source artifact, and boundary attached to each canonical claim.

Table 3: Progressive evidence chain from information characterization to evidence-calibrated decision realization.

Part	Stage	Scientific question	Key evidence	Effect	95% CI / domains	Narrative conclusion	Control boundary
Part I	I1 Spatial enrichment	Does spatial internal morphology enrich residual-capacity information?	Matched scalar and surface references versus selected B-family spatial field	MAE 0.18920 to 0.12849; reduction 0.06072 (32.1%)	familywise 95% CI [0.00664, 0.15364]; favorable 5/ 6 domains	Spatial morphology adds CAI-relevant information beyond lower-dimensional inputs.	The 0.08964 independent field estimate is sensitivity-only, not the matched confirmatory path.
	I2 Sparse recoverability	How much registered full-field information survives sparse observation?	Selected 25% bilinear sparse field versus surface and full-field references	89.9% gain retained; sparse-to-full MAE gap 0.00602	gap 95% CI [0.00173, 0.01083]; favorable 5/6 versus surface and 0/6 versus full	Sparse observations preserve most of the registered spatial gain.	Native-raster fraction is not physical scanner time and sparse is not equivalent to full field.
	I3 Spatial heterogeneity	Is task-relevant measurement opportunity spatially heterogeneous?	One-shot mechanical oracle versus uniform and reconstruction oracles	CAI-AUEBC improvements 0.00391 and 0.00373; 56.8% sequential headroom retained	95% CIs [0.00280, 0.00502] and [0.00284, 0.00469]; favorable 6/ 6	Specimen-specific spatial acquisition headroom exists retrospectively.	All oracle rows use unavailable counterfactual outcomes and are non-deployable.
	I4 Objective conditioning	Do downstream objectives induce distinct measurement priorities?	Mechanical and registered normalized-RGB-MSE reconstruction oracles plus learned global masks	CAI contrast 0.04862; image-MSE contrast 5.503e-4; learned indicator 0	95% CIs [0.04527, 0.05205] and [5.006e-4, 6.063e-4]	Retrospective priorities are objective-conditioned.	Learned global masks do not reproduce the oracle separation or establish deployment value.
	I5 State conditioning	Does measurement value evolve with inspection state?	Strict-OOF retrospective teacher from initial state to 18.75% checkpoint	70.4% best-action turnover; rank 0.405; top-5 overlap 0.307; opportunity 0.00531	descriptive across 276 specimens and 6 held-out domains	A fixed initial ranking omits material state-conditioned value evolution.	Teacher evolution does not show that the deployable scorer observes the change.

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Table 3 (continued)

Part	Stage	Scientific question	Key evidence	Effect	95% CI / domains	Narrative conclusion	Control boundary
	I6 Predictor conditioning	How does downstream predictor choice bound the task-value definition?	Ridge-Huber and Ridge-shallow-MLP strict-OOF value agreement	Spearman 0.762 versus 0.116; full-state MAE 0.08964, 0.08618, 0.15067	rank CIs [0.699, 0.821] and [0.069, 0.164]; 6 held-out domains	Value is stable between comparably performing low-complexity predictors.	The shallow MLP is substantially less accurate; equally accurate structurally distinct predictors remain unresolved.
Part II	II1 Static reference	What reference motivates state-conditioned valuation?	Static legal-state scorer with global and random exact-budget controls	static Spearman -0.0196; set regret 0.08171 versus 0.07993 and 0.07978	rank 95% CI [-0.0591, 0.0195]; favorable 3/6 domains	The static reference motivates conditional valuation.	This tested representation does not imply information-theoretic impossibility.
	II2 Dynamic valuation	Does state-conditioned scoring capture incremental dependence?	Dynamic real-state scorer versus static scorer at 18.75%	dynamic minus static regret -0.001260	95% CI [-0.002123, -0.000444]; favorable 5/6 domains	Dynamic valuation improves over the static reference.	The comparison alone does not identify which state-information source drives the gain.
	II3 Information-source attribution	Which legal-state information licenses content attribution?	Real content versus initial, acquired-position/history, reconstruction, and shuffled controls	real change -0.000731; real-minus-history 0.01740; real-minus-reconstruction 0.03419; real-minus-shuffled 2.328e-4	all matched-control CIs exclude 0; real favorable 1/6 for each source control	Real measurements change prediction, while matched controls localize the decision signal.	Acquired-position/history is not geometry-only; reconstruction is registered normalized-RGB-MSE; shuffled remains the dynamic boundary.
	II4 Valuation/planning decomposition	Where does controlled component headroom reside?	Privileged valuation and bounded planning substitutions	valuation 4.979e-5; learned planning 3.164e-6; true-value planning 1.117e-4	95% CIs [1.845e-5, 8.105e-5], [1.411e-7, 6.250e-6], [9.898e-5, 1.248e-4]	Valuation and planning contribute distinguishable retrospective headroom.	Substitutions are diagnostic and non-deployable; no representation bottleneck is claimed.

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Table 3 (continued)

Part	Stage	Scientific question	Key evidence	Effect	95% CI / domains	Narrative conclusion	Control boundary
	II5 Bounded set realization	How much decision-level headroom remains in the reachable pool?	Greedy and beam-width-four selection versus retrospective joint near-oracle	greedy regret 1.207e-4; beam-4 point regret 1.127e-4	95% CIs [1.033e-4, 1.386e-4] and [9.485e-5, 1.308e-4]	A positive two-action set-realization gap remains.	The near-oracle is non-deployable and does not establish arbitrary-horizon regret.
	II6 Deployment calibration	How does frozen end-to-end stress testing calibrate readiness?	Feedback/ no-feedback stress test and strongest deployable outer reference	no-feedback advantage 1.496e-5; residual deployable gap 6.114e-5	feedback-benefit CI [-1.944e-5, -1.064e-5]; baseline-minus-learned CI [-8.461e-5, -3.777e-5]	The frozen endpoint quantifies the remaining repre sentation-to-decision gap.	The no-feedback reference is retained and the learned endpoint is not performance-superior.

Part I identifies structured task-relevant information rather than a universal location score. Spatial morphology, sparse retention, heterogeneous oracle opportunity, objective specificity, and evolving teacher values jointly define what could be valuable to acquire. The oracle, learned-mask, and predictor boundaries keep this construct tied to a downstream model, state, objective, and legal information set.

Part II then tests realization rather than assuming it. State conditioning improves the registered dynamic scorer over a static reference, but acquired-position/history, reconstruction, and shuffled controls determine the narrow content claim that is licensed. Component substitutions and reachable-set analysis expose opportunities in valuation and planning; feedback and the frozen outer comparison calibrate how much of that structure reaches a deployable trajectory. The combined engineering-informatics result is a progressive evidence contract from information structure to decision outcome, not an equivalence between retrospective opportunity and policy performance.

Transfer conditions beyond the present case study.. The framework can be instantiated in another sensing problem only with five operational elements: a defined downstream engineering endpoint and loss; a legal partial-observation state excluding future measurements and outcomes; a defensible retrospective or counterfactual target for marginal information value; matched controls separating measurement content from acquisition geometry and history; and an end-to-end cost-constrained decision metric distinct from local value prediction. These are methodological conditions, not proof of universal empirical transfer. Independent composite studies provide convergent motivation rather than validation: interlocked thin-ply experiments show that a scalar change in delamination extent and the corresponding CAI change need not be equivalent [16, 17], and hybrid carbon–glass experiments likewise distinguish impact-damage resistance from CAI tolerance [18]. Neither study validates the present framework.

Four limitations bound this interpretation. First, the empirical evidence covers the six related CFRP experimental domains in the present paired public data program; evidence for other inspection systems, materials, or damage mechanisms is not established. Empirical statements therefore refer to results across the six held-out experimental domains in the present data program. Second, cost is exact native-raster location count, not travel time or industrial inspection cost. Third, the utility definition is specific to the chosen CAI predictors and cannot be read as causal attribution of failure mechanisms to selected image regions. Fourth, the action hierarchy, state representation, scorer, and bounded planners are fixed implementations. Their non-superiority does not prove that task-driven adaptive ultrasound is impossible; it shows which evidential conditions remain unmet before such a claim is warranted.

6. Conclusions

Part I establishes structured task-relevant information for post-impact assessment. Spatial ultrasonic morphology reduced held-out-domain CAI-ratio error by 32.1%, the registered sparse condition retained 89.9% of that gain, and retrospective priorities varied by location, downstream objective, and inspection state. These results define a valuable acquisition target while retaining the oracle, learned-global-mask, scanner-time, and predictor-conditioning boundaries.

Part II provides an evidence-calibrated account of decision realization. Dynamic valuation improved over the static reference, and controlled substitutions exposed separate valuation and bounded planning headroom. Acquired-position/history, reconstruction, and shuffled controls restrict the measured-content attribution, while the no-feedback reference remained preferred under the frozen stress test. The final learned endpoint retained a residual AUEBC gap of 6.114×10^{-5} and is not performance-superior to the strongest deployable reference.

The engineering implication is that adaptive inspection should progress from characterizing task-relevant information to validating legal-state attribution and finally to frozen decision outcomes. In the present six-domain CFRP case, that sequence reveals strong spatial structure and concrete decision headroom without overstating deployment readiness. The conclusions remain bounded to native-raster acquisition cost, the tested predictors and state representation, the fixed action hierarchy and planners, and the available paired experimental domains.

Data and code availability

The paired experimental datasets analyzed in this study are public sources cited in Section 4. An anonymized reproducibility package accompanies the submission and contains the paper-specific canonical metrics, source hashes, figure and table sources, supplemental machine-readable records, and deterministic source archive. The public code repository and archival identifier will be declared after review so that the review manuscript remains identity-neutral.

Declaration of generative AI and AI-assisted technologies in the writing process

During preparation of this work, the authors used OpenAI Codex to assist with evidence auditing, deterministic analysis-code generation, LaTeX drafting, and language editing. The authors reviewed and validated all generated content against the original machine-readable artifacts and take full responsibility for the manuscript.

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